

Rolling Shutter vs Global Shutter

Experiment 4

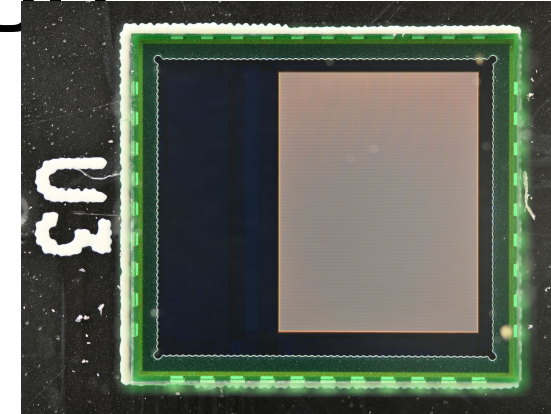
Sensors and Actuators Laboratory

Why does a rigid propeller look bent?

- Cameras do not necessarily capture every pixel at the same instant.
- A rapidly moving object may change position while the image is being acquired.
- The shutter mechanism determines the timing assigned to each image row.



What does a shutter do/control?



- A shutter controls when the sensor collects light.
 - Important terms:
 - T_{exp} = Exposure Time = How long each pixel/row collects light.
 - T_{line} = Line Time = Time offset between successive row exposures.
 - T_{skew} = Sensor scan time = Exposure-time difference between the first and last rows.
- $$T_{skew} \approx (N_{rows} - 1)T_{line}$$
- T_{frame} = Frame Interval = Time between successive completed frames.
 - Short T_{exp} reduces motion blur;
 - Short T_{skew} reduces rolling-shutter distortion.
 - Global shutter makes $T_{skew} \approx 0$. Stored image may still require time to read out

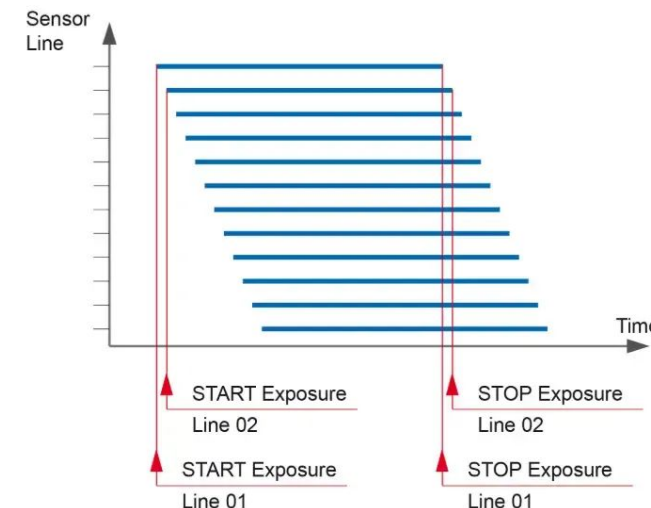
Global Shutter

- All pixels begin exposure together.
- All pixels stop exposure together.
- Charge is stored, then transferred to the output.
- The frame represents approximately one common time interval.
- Pixel values can be stored temporarily and read from the sensor afterward.

Read/Understand these links: [1](#) & [2](#) & [3](#)

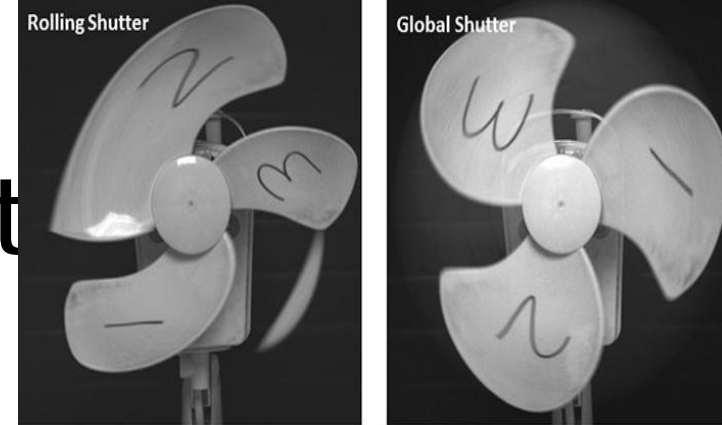
Rolling Shutter

- Builds the frame row-by-row.
- Rows are exposed at slightly different times.
- The exposure interval moves across the sensor.
- Each row may have the same exposure duration.
- The rows do not represent the same instant.
- Scan direction depends on the sensor architecture.



Read/Understand these links: [1](#) & [2](#) & [3](#)

Rolling Shutter Artifact



- Skew: Vertical structures lean when the camera or object moves horizontally.
- Wobble: Camera vibration causes different rows to experience different camera orientations.
- Partial Exposure: A short flash may illuminate only some rows.
- Distorted rotation: Propellers, fans, and wheel can appear curved, oddly shaped or disconnected.

Another link: [1](#)

Timing Comparison

Property	Global Shutter	Rolling Shutter
Start of exposure	Same for all rows	Delayed row by row
End of exposure	Same for all rows	Delayed row by row
Frame corresponds to	One common interval	A sequence of intervals
Moving-object geometry	Usually preserved	May be distorted
Typical implementation	Additional pixel storage	Simpler sequential timing

Comparison

Design Issue	Global Shutter	Rolling Shutter
Pixel Circuit	Storage and extra switching	Simpler
Motion Geometry	Preserved	Can distort
Sensitivity/Noise	Architecture-dependent trade-off	Favourable
Power and Cost	Higher	Lower
Modern Progress	BSI and stacked designs reduce penalties	Faster scans

Where is each shutter used?

- Rolling Shutter

- Smartphones
- Web cameras
- Consumer cameras
- General Photography
- Scenes with limited motion
- Applications prioritizing resolution, sensitivity, and cost

- Global Shutter

- Machine vision
- Industrial inspection
- Robotics with rapid motion
- Motion capture
- High-speed measurement
- Barcode and license-plate imaging
- Structured-light and stereo systems

Links: [1](#)

Final Remarks

- Rolling Shutter: One frame, many row times.
- Global Shutter: One exposure window for the whole frame.